Journal of Transportation Security (2026) 19:42
https://doi.org/10.1007/s12198-026-00363-w

RESEARCH

Danger by design? Mapping safe route choices in the city of Tshwane, South Africa

Magdelie Purchase[1] · **Gregory D. Breetzke**[1]

Received: 11 March 2026 / Accepted: 11 July 2026
© The Author(s) 2026

Abstract

Urban navigation systems typically prioritize the shortest or fastest routes between two points but rarely consider personal safety, a critical concern in cities characterised by elevated crime risks. In this study we delineate the safest and shortest route between any two locations in the city of Tshwane, South Africa. In a departure from previous work, we identify a safe route based on a combination of historical incivility data, the spatial distribution of various 'risky' facilities, as well as the location of numerous safety infrastructures such as surveillance cameras and security patrols. Results reveal substantial differences between the shortest route and various temporally-disaggregated safe routes, with the safest route being approximately 63% longer but exposing travellers to nearly half the number of recorded incivilities. Daytime, night-time, weekday and weekend routes also exhibited distinct patterns reflecting the routine activities of area users. The results offer practical implications for evidence-based safety interventions and the development of risk-aware navigation systems.

Keywords Safe route analysis · Crime risk mapping · Environmental criminology · Guardianship

✉ Gregory D. Breetzke
greg.breetzke@up.ac.za

Magdelie Purchase
u20458704@tuks.co.za

[1] Department of Geography, Geoinformatics and Meteorology, University of Pretoria, Pretoria 0002, South Africa

Published online: 25 July 2026

Introduction

Determining the fastest route between two known locations has become an essential feature of everyday technology. Navigation applications such as Google Maps and Apple Maps are well-known globally for providing users with options for finding the fastest routes between a specified starting and destination point by way of different transportation modes such as walking, or driving. Another well-known routing application, Waze, improved on these capabilities by incorporating real-time data entered by the users which pinpoints the locations of accidents, roadblocks, and/or dangerous driving conditions, among others. Importantly, these applications do not explicitly consider additional preferences of the traveller such as scenic quality, comfort, as well as personal safety and security. The latter is a particularly important omission as safety and security are central considerations that have been found to shape route choice, particularly in urban environments characterised by high crime risk (Amirgholy et al. 2017; Ghoul et al. 2023; Lusk et al. 2019). But what determines whether a particular route is 'safe' or not? Previous researchers have operationalised route safety in a number of ways, notably through the use of various crime-based measures (Galbrun et al. 2016; Goel et al. 2017; Mata et al. 2016; Puthige et al. 2021), and/or environmental risk indicators such as street lighting levels or traffic volume (Gershteyn and Terekhov 2024; Krumm & Horvitz et al., 2017; Nerella 2024). Another characteristic that has been applied in route-based safety assessments is the incorporation of various 'risky' facilities along the route being traversed. That is, the proximity along the route to facilities that have empirically been shown to be associated with elevated crime risk - such as alcohol outlets (Britt et al. 2005; Day et al. 2012), schools (Hewitt et al. 2018; Osborne et al. 2016), and transport interchanges (Irvin-Erickson and La Vigne 2015; Newton et al. 2015), among others. According to Eck et al. (2007), 'risky' facilities (often called crime generators or crime attractors) are places whose routine activities increase opportunities for crime by bringing together potential offenders and suitable targets in the absence of capable guardianship. One measure that has, however, rarely been integrated into route safety operations is the presence of guardianship or activity measures that may exist along transport routes. These infrastructures would *reduce* the perceived risk of crime along a particular route and can include CCTV cameras, and/or security patrols, among others. Drawing from routine activity theory (Cohen and Felson 1979), such features may reduce crime risk not by removing motivated offenders or suitable targets per se, but by increasing the likelihood of supervision and/or intervention. The notion is that routes with higher levels of guardianship are classified as being safer, all things considered.

In this study we add to what has been done in this scholarly space by determining the safest route between an origin and destination point within South Africa. In a departure from previous work, we identify a safe route based on a combination of historical crime (or incivility) data (increased risk), the spatial distribution of various 'risky' facilities (increased risk), *as well as* the location of numerous safety infrastructures such as surveillance cameras and security patrols (reduced risk). To our knowledge, this study is the first of its kind to incorporate the location of both crime *and* physical safety measures into a spatial routing model in the Global South. While much of the existing research on routing has been concentrated in the Global North

(see for example, Drawve and Barnum 2018; Kim et al. 2014), this study shifts the focus to the Global South, offering insights into the unique urban dynamics of South African cities, exploring how crime patterning and safe route identification may differ from northern contexts.

The geographical focus area for the study is the suburb of Hatfield located within the city of Tshwane in South Africa. More specifically, the Hatfield Community Improvement District (CID) which is an area of roughly 1 km² located within the suburb of Hatfield (see Fig. 1). A CID in South Africa is a geographically defined area within which property owners pay a levy for services and improvements to the area beyond those provided by the local municipality. These services include, among others, street cleaning, waste management, the maintenance of public spaces, and, importantly, enhanced security and policing. A central defining feature of the Hatfield CID is the main campus of the University of Pretoria - one of the largest public universities in South Africa. The university also plays a central role in shaping the demographic profile of the area manifested through a high concentration of students and student-oriented facilities such as bars, and restaurants alongside extensive private student accommodation and major transport interchanges. This mix of land uses, high-density population, and constant influx of mostly young people contributes to elevated opportunities for crime making identifying safe routes within this area a priority not only for students and staff, but also for residents, visitors, and the CID managers responsible for safety interventions.

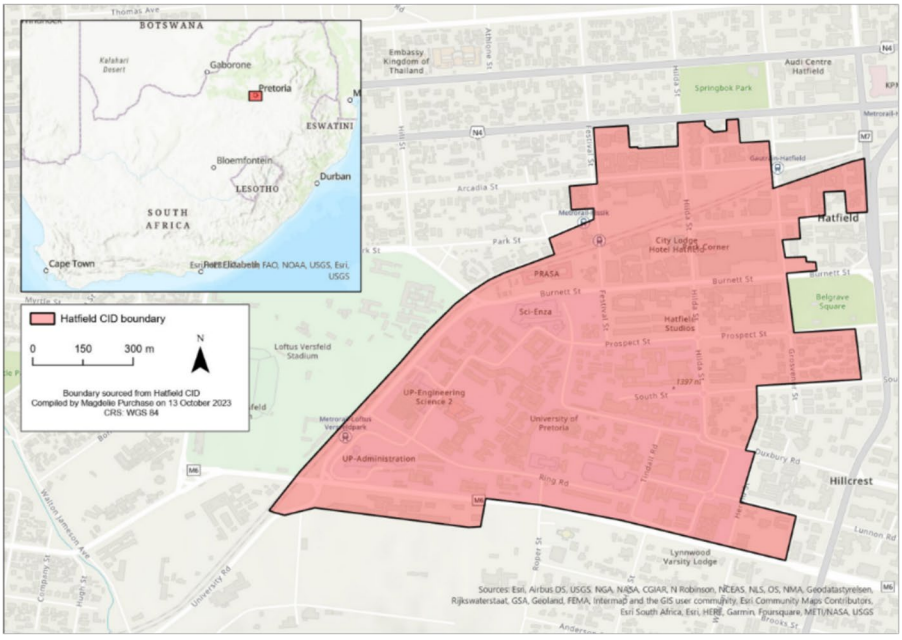


Fig. 1 The boundary of the Hatfield Community Improvement District (CID)

Literature review

A large number of researchers have considered the safety of the traveller when determining the safest route between two locations (see Hossain et al. 2022; Mata et al. 2016; Shah et al. 2011). The safety of a route is most often represented by some form of objective 'risk index' which assigns a numerical level of 'danger' to a particular street segment. Concomitantly, the safest (and shortest) routes are those that minimize traversing on streets with the highest risk index values whilst still getting to the desired destination in the shortest distance possible. Most often these indices are created using historical open crime datasets (see Galbrun et al. 2016; Goel et al. 2017; Mata et al. 2016; Puthige et al. 2021), although crowdsourced crime data has also been increasingly used (see Mata et al. 2016; Shah et al. 2011). In terms of the latter, Shah et al. (2011) developed a safety routing system (CROWDSAFE) in metropolitan Washington DC using data collected from crime reports submitted by individuals online while Kim et al. (2014) used sentiments inferred from real-time, geotagged messages from X (formerly Twitter) to delineate safer transport routes for users in Chicago. Other researchers have combined both historical and crowdsourced data to develop algorithmic approaches for identifying safer routes. These include Hossain et al. (2022), who developed the Safe Path for Everyone (SPaFE) algorithm, which integrated historical crime data with crowdsourced safety perceptions to generate route recommendations that minimised exposure to unsafe locations. The researchers developed the algorithm using four safety attributes (i.e., gender, severity, age, and time) in order to tailor route safety to different commuter vulnerabilities. Zaoad et al. (2023) advanced this model - renamed CrowdSPaFE - by incorporating historical crime data as an additional safety attribute alongside the original four, and by weighting different crime types according to their severity or associated sentencing. This allowed CrowdSPaFE to assess route safety not only in terms of demographic and temporal vulnerability but also in terms of actual crime risk exposure.

Other non-crime datasets have also been used to assign a level of 'danger' per street segment. For example, in the US, Sarraf and McGuire (2020) developed a Safe Route Planner navigation system utilising data on past motor vehicle crashes to provide the users with the safest route between two specified points while, also in the US, Pang et al. (2019) used street camera video data to identify the safety status of streets when designating a safe route. Earlier, Alfonso (2017) developed a safe routing application in California using various road barriers that act as impediments to travel such as intersections, walkways, freeways, and areas that had previously experienced pedestrian collisions while Eckdale-Dudley et al. (2018) also took infrastructural features such as crosswalks and sidewalks, and adequate traffic signage into account while designed a safe routing application in Wisconsin. Puthige et al. (2021) created a safest route detection system in New York using historical crime data, accident data, and road segment length. Their proposed solution aimed to provide end users with a warning regarding which routes are more dangerous than others, thereby giving them the opportunity to select their preferable route. The system did this by assigning street segments with a danger index which is a combined result of a crime score (weighted sum of crime events), an accident score (weighted sum of accidents) as well as the length (in kilometres) of a path. Routes with a lower total danger index

were seen as safer and more preferable. Finally, Oh et al. (2017) used both historical crime data and the location of various types of facilities when determining the safest and shortest route between two locations. In their study, the researchers calculated a risk rate for a route based on the crime risk rate of the facilities included in the route of the user. Each facility was assigned a crime risk value based on five crime types typically occurring at these facilities with the risk values ranging from one to ten with one considered the safest and ten the most unsafe. The routes were assigned crime risk values based on the facilities found along the way, and subsequently the routes which traversed more 'risky' facilities were deemed more unsafe as they had a higher likelihood of crime occurring.

Researchers have also identified the safest routes specifically for pedestrians. For example, Galbrun et al. (2016) used publicly accessible crime data to map the shortest pedestrian route between two specified points in Philadelphia and Chicago, weighting street segments by crime incidence, while Alfonso (2017) developed a route analysis solution known as Safe Routes to School (SRTS), which focused on identifying safer walking routes for children travelling to and from school. Using two elementary schools in California as case studies, the SRTS model integrated crime data with built-environment characteristics such as traffic volume, road crossings, sidewalk availability, and street connectivity. Finally, Eckdale-Dudley et al. (2018) designed a safe routing application in Wauwatosa, Wisconsin. The researchers created a heat map based on the locations of the student homes, in order to determine the areas which required safe routes for the students walking to school. The safe pedestrian routes were generated based on the location of the homes of students, existing walkways and the safety scores of the street crossings. The routes were made accessible to the parents of the students via an online application which provided the parents with widgets to search for the nearest safety route and also to provide the user with the subsequent directions for the chosen route.

Despite this plethora of past research, studies that have explicitly incorporated infrastructural features and/or facilities that increase perceived safety in safe route determination has, however, been less forthcoming. Guardianship in this instance refers to the presence of formal or informal protective elements along a street network that may reduce opportunities for crime and/or enhance perceived safety. Formal guardianship can include active surveillance by CCTV or security personnel or police, while informal elements could include the presence of well-maintained public lighting, or high pedestrian traffic that promotes natural surveillance. Just as certain characteristics can increase the perceived risk of traversing a particular route (high crime), so too can certain characteristics of the environment potentially reduce risk and collectively contribute to safer pedestrian routes. Our study aims to delineate the safest and shortest route between two locations considering historical crime data as well as the presence of 'risky' facilities (see Eck et al. 2007) along street segments but also, importantly, considering the presence of capable guardianship along the street network.

Data and methods

Incivilities data

Historical incivilities data were obtained from the Hatfield CID. The Hatfield CID team manually collect and record all incivilities that occur within the Hatfield CID boundary. A total of 580 point-level incivility incidents were captured for the Hatfield CID in 2024. Included in the dataset was a unique identifier per incident, the location of the incident, the type of incident, as well as a short description. Incivilities data were used instead of reported crime data because this type of data captures a broader range of perceived safety issues, including unreported and/or minor offenses, which provide a more comprehensive view of the safety of the area. According to Innes (2004) incivilities are called 'signal' crimes because they send powerful signals to the public about disorder and declining social control, thereby shaping perceptions of risk and fear beyond their immediate harm. The incivilities captured by the Hatfield CID team encompass a wider range of social and physical disorders that might not rise to the level of criminal offenses but still significantly impact community perceptions of safety and neighbourhood quality. Incivilities captured include fighting, littering, vandalism, public drinking, drug-related incidents, and/or pedestrian accidents. Reported crime data also has a number of limitations including underreporting due to factors like distrust in law enforcement, fear of retaliation, or cultural barriers. Incivilities data, on the other hand, can provide a more comprehensive picture of the social environment and has been closely linked to community perceptions of safety and quality of life in a neighbourhood in previous studies (see LaGrange et al. 1992; Taylor 2001).

Street network data

Street data was sourced from OpenStreetMap (OSM). The OSM network dataset follows a structured graph model where roads are represented using nodes (intersections) and ways (road segments), which ensures proper topological connectivity. The dataset was clipped to the Hatfield CID boundary and planarized to re-segment streets at each intersection, resulting in 1,318 street segments across the Hatfield CID. Attributes included road class, speed, direction, and segment length, with the latter being critical for route analysis.

Facility data

A total of 28 facilities known to act as crime generators were mapped to assess risk. These included two shopping malls, eight transport interchanges, ten alcohol outlets, four parks, and four educational institutions (one senior school and three tertiary facilities). The selection of these facilities was informed by both local (Breetzke and Edelstein 2022; Breetzke et al. 2021; Lancaster and Kamman 2016) and international studies (Day et al. 2012; Groff & McCord 2012; Newton et al. 2015; Pridemore and Grubesic 2013) which have previously found these types of facilities to be spatially associated with an increase in crime in their surrounding areas. Data was sourced

from OSM and the Tshwane Data Geoportal (https://opendata.tshwane.gov.za/app-default) - an online platform that provides spatial datasets, maps, and geospatial information for the **city** - with all locations manually verified using Google Maps.

Guardianship data

The Hatfield CID has a number of technological assets that are used to provide 'guardianship' across the area. These include fourteen CCTV cameras, and eighteen License Plate Recognition cameras. Moreover, stationary guards are located at various checkpoints throughout the CID at pre-determined locations and a series of mobile patrol guards (on foot and bicycles) patrol the CID on pre-determined routes. The locations of all the cameras, check points, and the details of the routes patrolled by the CID guards were provided by the Hatfield CID and digitized to point shapefile format. Importantly, for the mobile patrol guards, a stationary location or 'rest point' was provided which indicates the location where guards are most likely stationary when they are not actively patrolling their given route.

Analysis

Initially, an origin point was selected from one of the main campus gates at the University of Pretoria (specifically where students could exit campus), and the destination point was randomly selected within the Hatfield CID. The method to determine the safest (and shortest) route between these two points entailed several steps. First, a 30-metre buffer was created around each incivility point as well as around each of the 'risky' facilities in order to represent their relative influence on safety within its immediate vicinity. Street segments that intersected these buffered areas were considered 'dangerous' zones. Similarly, a buffer distance of 100-meters was created around each guardianship measure (i.e., cameras and guards). The buffer distance of 100 metres was used to indicate the increased relative influence of guardianship on crime risk reduction. If a street segment intersected the 100-metre buffer area, it was considered to be in a 'safe' zone. It is readily acknowledged here that the buffer distances used in the study (i.e., 30 metres and 100 metres respectively) are somewhat arbitrary. There is a distinct lack of scholarly contributions in criminology which provide any indication of the criminogenic effect of a 'risky facility' in distance as well as the protective effects of various crime prevention mechanisms. These distances were, therefore, assumed but we are reasonably confident that they do approximate their causative and reduction effect on crime, all things considered. Similarly, all incivility incidents were assigned the same 30 metre buffer irrespective of incident type. While it could be that certain incivilities may exert a broader spatial 'influence' on perceived and/or actual risk than others, there is currently little empirical evidence to guide the assignment of incident-specific influence distances. Applying a uniform buffer distance, therefore, ensured methodological consistency and avoided introducing subjective assumptions into the routing model.

Next, these 'dangerous' and 'safe' zones were used as route polygon barriers for the routing analysis. Barriers were ascribed as being either at a restriction or a scaled cost. A barrier is a restriction if it has to be avoided completely when choosing a

route while a barrier is a scaled cost if it merely altered the cost of travel when being traversed. In this study, all 'dangerous' zones were set at a scaled cost meaning that the cost of travel increased if a street segment intersected these zones. All scale costs were set to the same value ($n = 1.5$) for all historical incivilities and facilities. This means that no 'dangerous' zone was considered 'more criminogenic' than another dangerous zone. Similarly, in order to apply an equal, but opposite, value to the 'safe' zones a scaled cost value of 0.5 was applied to street segments that traversed a 'safe' zone. If zones overlapped, for example, when incivilities or facilities are located close to each other; their individual scaled costs were multiplied to form a new combined weight. For example, if an incivility barrier overlapped with a transport interchange polygon barrier their weights were multiplied, and this multiplied value was applied to the portion of the street segment that passed through the overlapping polygon barriers. The resultant risk rates of the 'dangerous' and 'safe' zones along a particular route were combined in order to acquire the overall risk rate of the entire route. This methodology was applied across three temporal resolutions: overall, daytime/night-time, and weekday/weekend, based on the timing of incivility incidents. All other data included in the routing analysis – facilities and guardianship - were considered temporally stagnant for these iterations. That is, all incivility data were partitioned into daytime/night-time and weekday/weekend subsets, and separate routing analyses were undertaken for each subset. In contrast, the locations of risky facilities and guardianship measures were treated as temporally static because time-specific data on their operation and/or deployment (e.g., business hours and/or patrol schedules) were not consistently available. Each subset produced a unique route reflecting the spatial and temporal distribution of risk.

Results

Figure 2 shows the shortest route (in green) between the selected origin and destination points within the Hatfield CID. The origin location (in the western region of the CID) is one of the main entrances of the University of Pretoria, and the destination location is a random point (in the eastern region of the CID). Importantly, this route does not take any of the underlying environmental backcloth into account such as the various 'dangerous' and 'safe' zones and is therefore simply the shortest route between these two points.

Figure 3 shows the safest route (in yellow) between the same two points considering all of the historical incivilities, 'risky' facilities and guardianship encountered along the route. Immediately noticeable is the fact that this route is considerably longer than the shortest route. This route still traversed 41 historical incivilities as well as eight risky facilities.

Figure 4 shows the safest route (in purple) for the daytime hours (between 06:00am and 18:00pm) while Fig. 5 shows the safest route (in orange) for the night-time hours (between 18:00pm and 06:00am). These routes are differentiated based on the number of historical incivilities that occurred during the day, compared to the number of historical incivilities that occurred during the night respectively. The 'risky' facilities and safety measures are considered permanent in these iterations of the safest routes

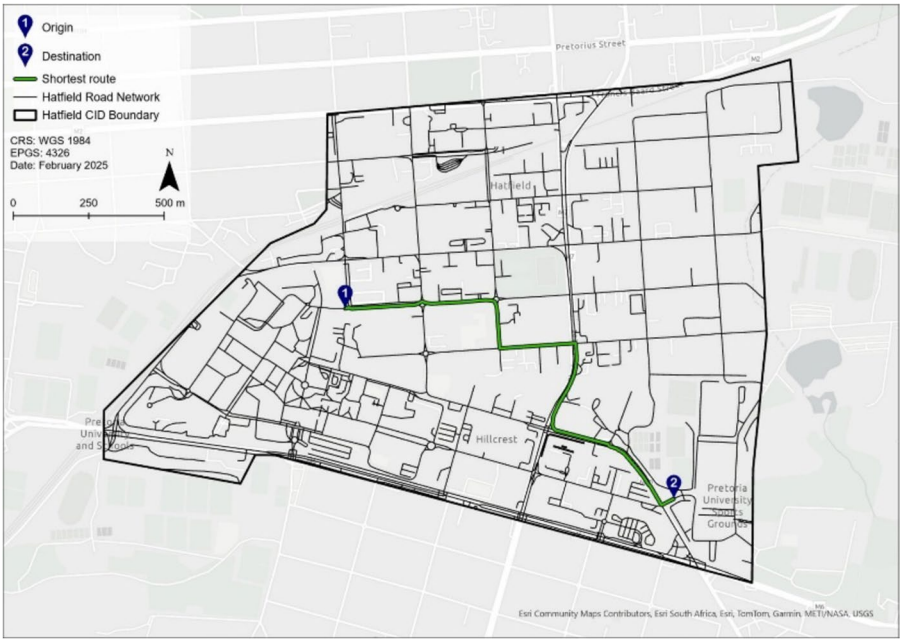


Fig. 2 The shortest route between the origin and destination point within the Hatfield CID



Fig. 3 The safest route between the origin and destination point within the Hatfield CID

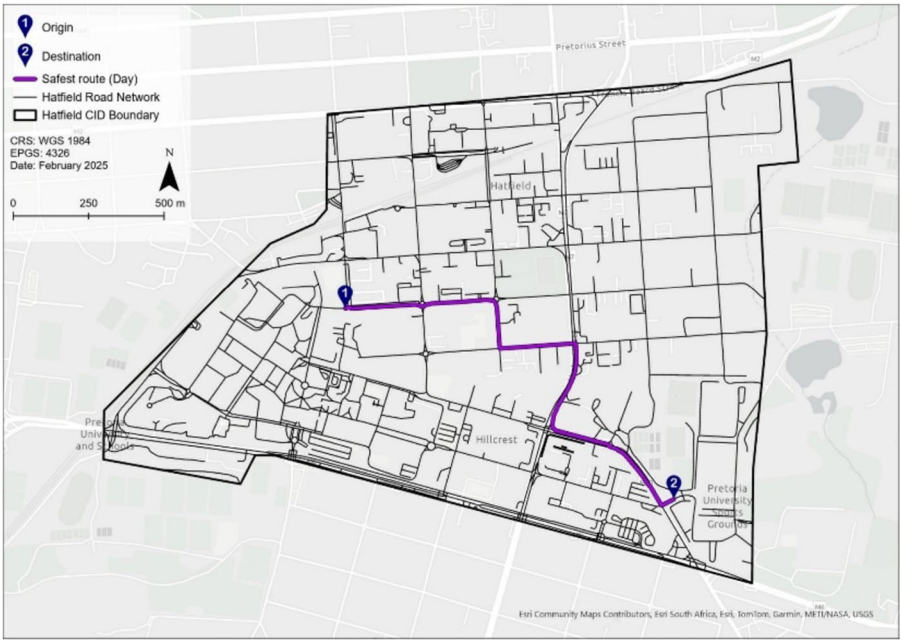


Fig. 4 The safest route during the day between the origin and destination point within the Hatfield CID



Fig. 5 The safest route during the night between the origin and destination point within the Hatfield CID

as these facilities and measures are considered temporally stagnant. Interestingly, the safest route for daytime hours is the same as the shortest route between the origin and destination points (see Fig. 2). The safest route for the day traversed 30 historical incivilities whereas the safest route for night traversed 25 historical incivilities.

Figure 6 shows the safest route for a weekday (in blue) and Fig. 7 indicates the safest route for the weekend (in red). Again, these routes are differentiated based on when the historical incivilities occurred (i.e., weekday versus weekend). The 'risky' facilities and safety measures are, again, considered permanent in these iterations of the safest routes. Both routes initially follow a similar path, but the weekend route is noticeably shorter than that of the weekday route. The weekday route traversed 39 historical incivilities, while the weekend route traversed 28 historical incivilities. The weekday route traversed 11 risky facilities, compared to seven risky facilities on the weekend route.

Figure 8 shows all the routes generated in the study from the origin to the destination points collectively. The safest route, the night-time route and the weekday route initially follow the same path through the Hatfield CID, with the weekday route and night-time route deviating towards the destination point. The weekday route deviates first to follow the same path as the day-time route and weekend route towards the destination point. Similarly, the day-time route and the weekend route follow the same path initially, only deviating from one another closer to the destination location. Furthermore, we can see that the safest route for the weekend is the only route which does not follow the same path as the other safety routes, when approaching the destination point. All five of these routes share some segments of the road network.



Fig. 6 The safest route during the week between the origin and destination point within the Hatfield CID

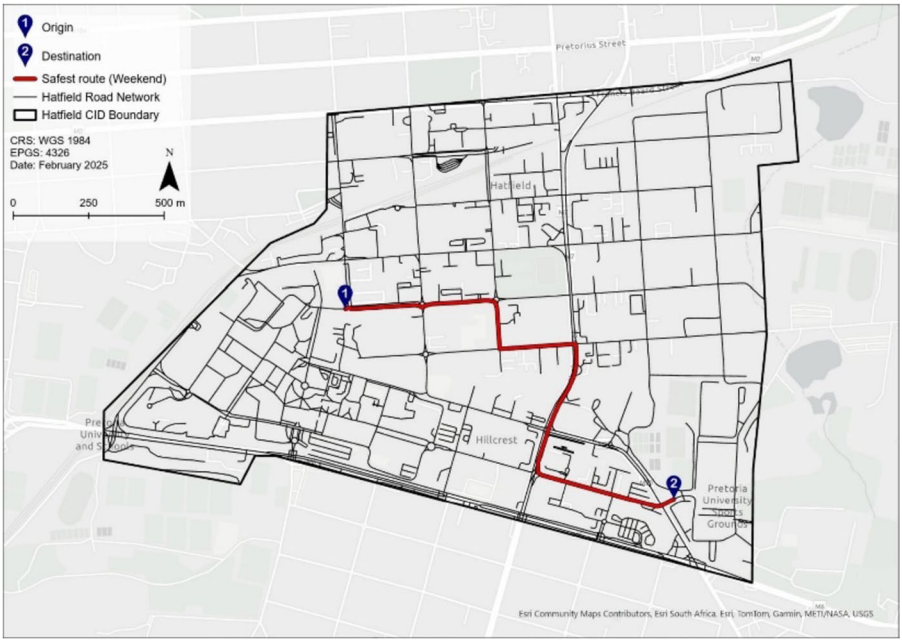


Fig. 7 The safest route during the weekend between the origin and destination point within the Hatfield CID

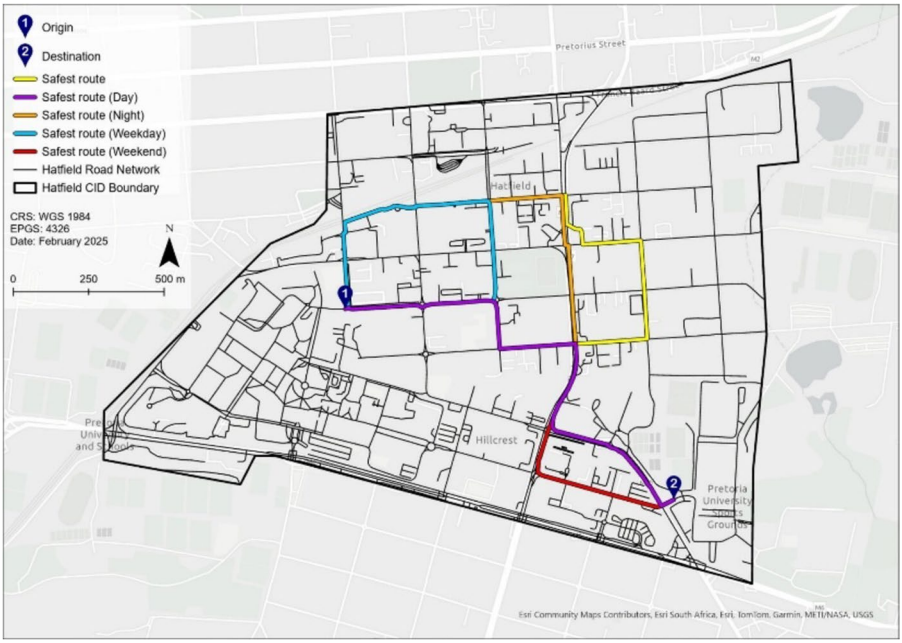


Fig. 8 The safest routes for all subsets of incivilities calculated within the Hatfield CID

Table 1 Comparison of the total length and measures encountered for each 'safe' route

Route	Length (m)	Incivilities (*n*)	Risky facilities (*n*)	Safe zones (*n*)
Shortest route	1722.91	81	7	11
Safest route	2814.33	41	8	10
Daytime route	1723.49	30	7	11
Night-time route	2351.79	25	9	8
Weekday route	2352.13	39	11	13
Weekend route	1836.95	28	7	12

Finally, Table 1 shows a comparison of the total length and number of danger and safety measures encountered for each 'safe' route. The daytime route was substantially shorter than the night time route but, interestingly, traversed more historical incivilities than the night time route. The weekday and weekend routes also differed significantly. That is, the weekday route was longer than the weekend route and traversed more historical incivilities and 'risky' facilities than the weekend route, respectively.

Discussion

In this study we determined the safest route between two points within the Hatfield CID in the city of Tshwane considering the presence of historical incivilities, the location of certain 'risky' facilities, as well as the presence of various forms of guardianship. Major differences were found between the shortest route, and the safest route - selected based on the above-mentioned criteria. First, the shortest route between the two identified points was considerably shorter (1722.91 m) than the safest route ($n = 2814.33$ m). Importantly, the shortest route passed 81 historical incivilities and seven 'risky' facilities along the route compared with 41 historical incivilities and eight 'risky' facilities in the safest route. Both also differed significantly in the direction taken along the streets of the Hatfield CID. This indicates that if students (residents) prioritise safety over efficiency, they would need to travel substantially longer distances but with reduced exposure to risk. This suggests a clear trade-off between route directness and perceived or actual safety. While the safest route is approximately 63% longer than the shortest option, it exposes individuals to nearly half the number of recorded incivilities, highlighting the value of incorporating environmental risk indicators into route planning.

After disaggregating the incivility data and re-running the routing analysis, we, again, found a number of differences between the projected routes with the daytime route being substantially shorter than the night-time route despite traversing more historical incivilities, while the weekday route was longer and intersected more historical incivilities and 'risky' facilities than the weekend route. To some extent, these patterns reflect the routine activities of individuals within the Hatfield CID. More specifically, these differences point to shifting temporal routines and their influence on exposure to risk and the presence of guardianship. For example, during the week, predictable routines may allow would-offenders to anticipate pedestrian movement, while on weekends, leisure activities and alcohol consumption may increase vulner-

ability in certain areas of the Hatfield CID. Consequently, for students and residents within the Hatfield CID, especially those travelling during higher-risk periods, opting for slightly longer routes may significantly enhance personal safety despite the additional time and effort required. Previous studies have shown that crime patterns follow temporal rhythms (Andresen and Malleson 2015; Breetzke and Cohn 2012), reinforcing the importance of incorporating time-based analysis into spatial crime prevention strategies. Moreover, the study clearly demonstrated that the shortest route is not necessarily the safest (Amirgholy et al. 2017; Galbrun et al. 2016). While longer routes may, on average, reduce overall exposure to historical incivilities, they do not necessarily increase access to guardianship. There is a trade-off. For example, the night-time route is one of the longer routes (2351,79 m) yet traversed the lowest number of guardianship measures ($n = 8$) of all routes, indicating limited surveillance coverage. These findings highlight the need to balance distance, risk mitigation with safety accessibility, and to consider both spatial and temporal factors when planning pedestrian routes.

This study offers a number of practical implications for improving user safety that extend beyond technical routing. First, safe routes should be operationalized and available to residents (and students) within the Hatfield CID through user-friendly digital platforms. This will enable access to real-time, risk-aware navigation via smartphone-based applications. Such tools could integrate dynamic environmental risk indicators - including incivilities, risky facilities, and guardianship features - allowing users to make informed route choices based on both spatial and temporal conditions. In this way, safe routing moves from a purely theoretical exercise to a practical, everyday decision-support system that enhances situational awareness and actually supports safer mobility within the Hatfield CID. Second, additional safety/guardianship measures should be implemented along the high-risk route segments identified in our routing analysis. These represent locations where pedestrians may be exposed to greater risk and where targeted interventions could strengthen guardianship. This can be done through, among others, the provision of additional street lighting, or adding clear signage indicating which streets/routes are being patrolled and/or are under surveillance. Furthermore, the Hatfield CID team could determine the areas actively avoided by travellers and classify certain street segments as locations of higher risk to users. Patrol units could also be deployed to certain street segments based on time of day, and day of the week with the highest risk to users, in order to maximize the influence of increased guardianship and safety of travelers within the Hatfield CID. Third, the Hatfield CID employs dedicated maintenance teams responsible for the ongoing upkeep and improving the visual quality of the area, including waste removal, vegetation management, and the cleaning of graffiti and illegal advertising. Integrating routing data into their operations would allow maintenance efforts to be more strategically targeted toward streets/routes with higher risk levels and/or lower perceived safety. Existing research highlights the importance of perceived safety in shaping the functionality and use of urban environments (see Dubey et al. 2025; Jiang et al. 2018). Poorly maintained areas may signal a lack of oversight, thereby increasing feelings of vulnerability, whereas clean and well-maintained environments can foster a sense of surveillance and safety. Finally, ‘risky’ facilities could

be prioritised for regular maintenance to minimise incivilities and improve overall user experience.

Finally, there are some limitations to consider in this study. First, this study utilised incivility data captured by the Hatfield CID. As with any reported crime data there is a risk of underreporting. Furthermore, as the data is collected by the patrolling and security guards, the dataset might not include incidents occurring behind closed doors. Therefore, we can assume that the dataset compiled by the Hatfield CID team most likely does not include all incidents, both incivility and criminal, occurring within the Hatfield CID for the study period. Despite this, however, we believe that the dataset obtained from the Hatfield CID still provides a useful and contextually relevant representation of spatial patterns of incivilities within the Hatfield CID, allowing for meaningful analysis of risk and route safety. Second, all 'risky' facilities were treated as being uniform crime generators, irrespective of temporal variation. In reality, the criminogenic influence of such facilities is likely to fluctuate by time of day and day of the week. For example, a bar may present elevated risks during the weekend and/or during the night-time but far lower risks during weekday mornings. By not incorporating these temporal dynamics, the analysis may oversimplify the relationship between facilities and crime risk, potentially obscuring more nuanced patterns of exposure. The aim of this research was to provide a useful baseline representation of environmental risk that remains valuable for route planning and decision-making. Treating facilities as consistently risky ensured a more conservative estimate of exposure, which may be preferable in safety-oriented applications. Moreover, the framework developed in this study is inherently adaptable and can be refined in future work to incorporate temporal weighting of facilities as more detailed time-specific data become available, thereby enhancing the precision and applicability of the model. Third, this study is context specific and the safest routes determined are specific to the physical street network, historical incivilities and location of risky facilities in the Hatfield CID. Results will differ in different geographical contexts. Last, the temporal analyses considered only the timing of incivility incidents. Although facilities such as alcohol outlets and/or transport interchanges are likely to vary in their criminogenic influence according to their hours of operation, and guardianship may also fluctuate over time, these temporal dynamics were not incorporated because suitable time-specific data were not available to us.

These limitations notwithstanding, this study represents one of the first empirical attempts to incorporate measures that, perceptively, either increase or decrease the risk of traversing a particular route when delineating the shortest (and safest) route between two points. In doing so, it contributes to the growing body of scholarly work that maps safe routes, particularly within the Global South, where such research remains limited. By providing a replicable method for identifying safe routes, this study offers practical implications for the Hatfield CID, including supporting evidence-based allocation of resources, guiding the targeted deployment of security personnel and infrastructure, and informing the creation of designated safety corridors that can improve user mobility and reduce exposure to victimisation. Given the importance of overall safety not only within the Hatfield CID, but also across South Africa as a whole, it has become important for individuals to be adequately informed of their risk of victimisation at any given moment. We believe that this study can

hopefully contribute in a small way to the safety of the individuals within the Hatfield CID and can hopefully serve and the foundation of future policies, applications or initiatives aimed at further increasing the safety of individuals within the area.

Author contributions Magdelie Purchase did the spatial analysis for the project, and Gregory Breetzke supervised and wrote the main manuscript text. All authors reviewed the manuscript.

Funding Open access funding provided by University of Pretoria. No funding was received for this study.

Data availability No datasets were generated or analysed during the current study.

Declarations

Competing interests The authors declare no competing interests.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by/4.0/.

References

Alfonso JR (2017) Safe walk: A network analyst framework for safe routes to school. Unpublished Master's thesis, University of Southern California, United States

Amirgholy M, Golshani N, Schneider C, Gonzales EJ, Gao HO (2017) An advanced traveler navigation system adapted to route choice preferences of the individual users. Int J Transp Sci Technol 6(4):240–254. https://doi.org/10.1016/j.ijtst.2017.10.001

Andresen MA, Malleson N (2015) Intra-week spatial-temporal patterns of crime. Crime Sci 4(1):12

Breetzke GD, Cohn EG (2012) Seasonal assault and neighborhood deprivation in South Africa: some preliminary findings. Environ Behav 44(5):641–667. https://doi.org/10.1177/0013916510397758

Breetzke GD, Edelstein IS (2022) Do crime generators exist in a developing context? An exploratory study in the township of Khayelitsha, South Africa. Secur J 35(2):1–18. https://doi.org/10.1057/s41284-020-00264-0

Breetzke GD, Fabris-Rotelli I, Modiba J, Edelstein IS (2021) The proximity of sexual violence to schools: Evidence from a township in South Africa. GeoJournal 86:765–776. https://doi.org/10.1007/s10708-019-10093-3

Britt H, Carlin B, Toomey T (2005) Neighborhood level spatial analysis of the relationship between alcohol outlet density and criminal violence. Environ Ecol Stat 12(4):411–426

Cohen L, Felson M (1979) Social change and crime rate trends: A routine activity approach. Am Sociol Rev 44(4):588–608

Day P, Breetzke GD, Kingham S, Campbell M (2012) Close proximity to alcohol outlets is associated with increased serious violent crime in New Zealand. Aust N Z J Public Health 36(1):44–50. https://doi.org/10.1111/j.1753-6405.2012.00827.x

Drawve G, Barnum JD (2018) Place-based risk factors for aggravated assault across police divisions in Little Rock, Arkansas. J Crime Justice 41(2):173–192

Dubey S, Bailey A, Lee JB (2025) Women's perceived safety in public places and public transport: a narrative review of contributing factors and measurement methods. Cities 156:105534

Eck JE, Clarke RV, Guerette RT (2007) Risky facilities: Crime concentration in homogeneous sets of establishments and facilities. Crime Prev Stud 21:225–264

Eckdale-Dudley R, Sukupcak A, Engelking K, Simpson D (2018) A smarter, safer way to walk to school. https://www.esri.com/esri-news/arcnews/spring18articles/a-smarter-safer-way-to-walk-to-school. Accessed 14 August 2023

Galbrun E, Pelechrinis K, Terzi E (2016) Urban navigation beyond shortest route: The case of safe paths. Inform Syst 57:160–171. https://doi.org/10.1016/j.is.2015.10.005

Gershteyn AM, Terekhov AN (2024) A Simple Way to Improve Road Safety by Avoiding Dangerous Sections of the Route. Program Comput Soft 50(Suppl 2):S120–S127. https://doi.org/10.1134/S0361768824700506

Ghoul T, Sayed T, Fu C (2023) Real-time safest route identification: examining the trade-off between safest and fastest routes. Anal Methods Accid Res 39:100277. https://doi.org/10.1016/j.amar.2023.100277

Goel N, Sharma R, Nikhil N, Mahanoor S, Saini M (2017) A crowdsourced adaptive safe navigation for smart cities. In *2017 IEEE International Symposium on Multimedia (ISM)* (pp. 382–387). IEEE

Groff E, McCord ES (2012) The role of neighborhood parks as crime generators. Secur J 25:1–24. https://doi.org/10.1057/sj.2011.1

Hewitt A, Beauregard E, Andresen MA, Brantingham PL (2018) Identifying the nature of risky places for sexual crime: The applicability of crime pattern and social disorganization theories in a Canadian context. J Criminal Justice 57:35–46

Hossain E, Karim MR, Hasan M, Zaoad SA, Tanjim T, Khan MM (2022) SPaFE: A crowdsourcing and multimodal recommender system to ensure travel safety in a city. IEEE Access 10:71221–71232. https://doi.org/10.1109/ACCESS.2022.3187964

Innes M (2004) Signal crimes and signal disorders: Notes on deviance as communicative action 1. Br J Sociol 55(3):335–355

Irvin-Erickson Y, La Vigne N (2015) A spatio-temporal analysis of crime at Washington, DC metro rail: stations' crime-generating and crime attracting characteristics as transportation nodes and places. Crime Sci 4:1–13

Jiang B, Mak CNS, Zhong H, Larsen L, Webster CJ (2018) From broken windows to perceived routine activities: examining impacts of environmental interventions on perceived safety of urban alleys. Front Psychol 9:2450

Kim J, Cha M, Sandholm T (2014), April SocRoutes: safe routes based on tweet sentiments. In *Proceedings of the 23rd International Conference on World Wide Web* (pp. 179–182)

Krumm J, Horvitz E (2017) Risk-Aware Planning: Methods and Case Study on Safe Driving Routes. Proceedings of the Twenty-Ninth AAAI Conference on Innovative Applications (IAAI-17)), 4708–4714

LaGrange RL, Ferraro K, Supancic M (1992) Perceived risk and fear of crime: role of social and physical incivilities. J Res Crime Delinquency 29:311–334

Lancaster L, Kamman E (2016) Risky localities: Measuring socioeconomic characteristics of high murder areas. South Afr Crime Q 56:27–35

Lusk AC, Willett WC, Morris V, Byner C, Li Y (2019) Bicycle facilities safest from crime and crashes: Perceptions of residents familiar with higher crime/lower income neighborhoods in Boston. Int J Environ Res Public Health 16(3):484. https://doi.org/10.3390/ijerph16030484

Mata F, Torres-Ruiz M, Guzmán G, Quintero R, Zagal-Flores R, Moreno-Ibarra M, Loza E (2016) A mobile information system based on crowd-sensed and official crime data for finding safe routes: A case study of Mexico city. Mob Inform Syst. https://doi.org/10.1155/2016/8068209

Nerella T (2024) *Impact of Street Lighting on Pedestrian Path Routing: Impedance Factor Implementation* (Master's thesis, Georgia Institute of Technology)

Newton A, Partridge H, Gill A (2015) In and around: Identifying predictors of theft within and near to major mass underground transit systems. In: Ceccato V, Newton A (eds) Safety and Security in Transit Environments. Palgrave Macmillan, pp 99–115. https://doi.org/10.1057/9781137457653_6

Oh SB, Park HJ, Kang BK, Doo IC, Park MH (2017) Safe route recommendation method to prevent crime exposure. Adv Sci Lett 23(10):9579–9583. https://doi.org/10.1166/asl.2017.9751

Osborne D, McCord ES, Higgins GE (2016) The interactive influence of prosocial places, youth population, and poverty on incidents of violent assault. Deviant Behav 37(4):385–400

Pang Y, Zhang L, Ding H, Fang Y, Chen S (2019) SPATH: Finding the safest walking path in smart cities. IEEE Trans Veh Technol 68(7):7071–7079. https://doi.org/10.1109/TVT.2019.2918576

Pridemore WA, Grubesic TH (2013) Alcohol outlets and community levels of interpersonal violence: Spatial density, outlet type, and seriousness of assault. J Res Crime Delinquency 50(1):132–159. https://doi.org/10.1177/0022427810397952

Puthige I, Bansal K, Bindra C, Kapur M, Singh D, Mishra VK, Aggarwal A, Lee J, Kang BG, Nam Y, Mostafa RR (2021) Safest route detection via danger index calculation and k-means clustering. Computers Mater Continua 69(2):2761–2777

Sarraf R, McGuire MP (2020) Integration and comparison of multi-criteria decision-making methods in safe route planner. Expert Syst Appl 154:113399. https://doi.org/10.1016/j.eswa.2020.113399

Shah S, Bao F, Lu C-T, Chen I-R (2011) CROWDSAFE: Crowd sourcing of crime incidents and safe routing on mobile devices. In *Proceedings of the 19th ACM SIGSPATIAL International Conference on Advances in Geographic Information Systems (GIS '11)* (pp. 521–524)

Taylor RB (2001) Breaking away from broken windows: Baltimore neighborhoods and the nationwide fight against crime, grime, fear, and decline. Westview Press, Boulder, CO

Zaoad SA, Mamun-Or-Rashid M, Khan M (2023) CrowdSPaFE: A crowd-sourced multimodal recommendation system for urban route safety. IEEE Access 11:23157–23166. https://doi.org/10.1109/ACCESS.2023.3252881

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.